MILITARY SPECIFICATION

PROJECTILE, 155MM: XM825 METAL PARTS ASSEMBLY

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. COPE

1.1 This specification covers Projectile, 155MM, XM825 Metal Parts.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-6867 - Magnetic Inspection Units
MIL-I-6868 - Inspection Process, Magnetic Particle
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions For

STANDARDS

MILITARY

MIL-STD-105 - Sampling rocedures and Tables for Inspection by Attributes
MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Sampling by Attributes

FSC 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

9352634 - Projectile, 155MM, XM825, Metal Parts
8837839 - Pallet for Projectile, 155MM, HE and GB Hollow Base Type

INSPECTION EQUIPMENT DRAWINGS

9280318 - Body Magnetic Particle Standard for 155MM, ICM, M483A1
T9280354 - Eddy Current Standard, XM825 Base
T9280355 - Envelope Drawing Eddy Current Equipment, XM825 Base

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS

AMS 2375 - Control of Forgings Requiring First Article Approval

(Applications for copies should be addressed to: Society of Automotive Engineers, 400 Commonwealth Drive, Warrandale, PA 15906).

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM Method E-8 - Tension Testing of Metallic Materials
ASTM Method E-10 - Brinell Hardness of Metallic Materials
ASTM Method E-103 - Rapid Indentation Hardness Testing of Metallic Materials
ASTM Method B-117 - Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and standards.

3.2 Body.

3.2.1 Forming. When a cold working operation is used in forming the body, the maximum reduction in cross sectional area shall not exceed 22 percent. Process drawings showing the prior to cold work and after cold work process piece dimensions shall be submitted to the technical agency for approval prior to use. Once approved, the process drawings shall not be changed without prior approval of the procuring activity. The hardness of the process piece prior to a cold working operation shall not exceed 241 Brinell (see 6.6).

3.2.2 Stress relief. Prior to the band welding operation, bodies which have been cold worked shall be stress relieved at 1000°F (538°C) minimum temperature (body) for a minimum time of 30 minutes.

3.2.3 Heat treatment. Each body (after application of the welded overlay band) shall be austenitized, quenched and tempered uniformly to meet the mechanical properties specified on the applicable drawing. No work involving plastic flow is permitted after tempering unless followed by a stress relief. (see 6.12).

3.3 Rotating band.

3.3.1 Rotating band application. The rotating band shall be overlay welded using the gas metal arc welding process in accordance with the applicable drawings and specifications. Gas metal arc welding process in this instance means the use of a consumable electrode with a non-electrode auxiliary wire. The width of the band shall be obtained by oscillating the welding head and the auxiliary wire feed system. The process for applying the rotating band, once established must be maintained, controlled and monitored by the contractor. Procedures for maintaining, controlling and monitoring the process must be submitted to the technical agency for approval prior to the start of production. Documented records of the process control must be made available to the Government upon request (see 6.9).

3.3.2 Rotating band seat. Body material (steel) under the rotating band shall not be visible in the cannelure diameter.

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3.3.3 Foreign matter on band seat. Prior to banding, the band seat shall be free of any contaminants that would be detrimental to the adhesion of the band to the band seat.

3.4 Ogive and base.

3.4.1 Microstructure. The microstructure of the ogive and base shall comply with the microstructure depicted in Figure 1a or 1b.

3.4.2 Heat treatment. Each ogive and base shall be solution heat treated and aged uniformly to meet the mechanical properties specified on the applicable drawing.

3.4.3 Monitoring of hot forge temperature. When ogives and bases are hot forged, the forger shall provide continuous, automatic monitoring of slug temperature prior to hot forging. Details shall be provided in accordance with paragraph 3.8.1.

3.4.4 Storage. Forgings, semi finished, or finished components shall be stored indoors. Forgings shall be processed into finished components in a sequential forging lot basis; i.e., first in first out.

3.5 Metal defects.

3.5.1 Body, ogive and base. The body, ogive and base shall be free from all metal defects, in accordance with the defect standards (see 3.5.3) including cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams and other metal defects. In addition, the cavity of the components shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, and pits (see 3.5.3).

3.5.2 Rotating band. The rotating band and band seat areas shall have no cracks, lack of bonding, or other metal defects in accordance with defect standards (see 3.5.3).

3.5.3 Defect standards. Defect standards must be approved by the technical engineering agency. All visual standards for defects will be established for each contract after the first months' production or whenever requested by the procuring contracting agency.

3.6 Heat treatment of base adapter. Each base adapter shall be austenitized, quenched and tempered uniformily to meet the mechanical proterties specified on the applicable drawing. No work involving plastic flow is permitted after tempering unless followed by a stress relief (see 6.12).

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3.7 Proving Ground Function Test. The metal parts when assembled to form the projectile, 155MM, XM825 shall reveal no evidence of failure when tested at excess and service pressure or Zone 3. Metal parts failure shall be:

a. Loss or breakup of metal part components in flight (except obturator).
b. Shear of the rotating band as evidenced by complete lack of engraving.
c. Base and cargo not ejecting (except excess pressure) (Failure analysis must determine the cause to be the projectile metal parts).

3.8 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.8.1 First article approval of aluminum forgings. First article approval of forgings, forging drawings, process and controls to be used in producing subsequent production forgings shall be obtained from the government technical agency prior to initiating production. The procedures for obtaining first article approval shall be in accordance with AMS 2375.

3.9 Records of results of contractor conducted examinations and tests. The contractor shall maintain records of the results of all examinations and tests required by this specification and submit same upon request by the Government (see 6.4).

3.10 Workmanship. All parts and assemblies shall be free of chips, burrs, sharp edges, dirt, grease, oil, rust, or other foreign matter. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.11 Painting. Painting shall comply with the requirements of the applicable drawing. Drying Time prior to testing shall be in accordance with the applicable drawing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities as indicated.

PART DESCRIPTION	DRAWINGS	QUANTITY
Projectile Metal Parts Assembly, complete with base and adapter assembly	9352634	25
Projectile subassembly	9352633	5
Body	9352632	5
Body (Prior to Banding)	9352632	2
Ogive	10542916	5
Ogive Forging (Prior to heat treat)		1
Base and adapter assembly	9352631	25
Base	9352629	5
Base (Prior to protective finish and drilling wrench slots)	9352629	5
Base Forging (Prior to heat treat)	9532629	1
Adapter, Base	9352630	10
Obtuarator	10542907	5

Twenty of the above projectile metal parts assemblies shall be forwarded to a designated load plant for loading and shall then be forwarded to a Government proving ground for ballistic testing.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table 1 herein.

4.3.3 Rejection. See MIL-A-48078.

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Table 1. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
-	Metal Parts Assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ogive (Dwg. 10542916)				
	Examination for Defects	5	-	3.1	4.4.2.1
	Protective finish test	1	-	3.1	4.5.10
	Yield strength & Percent Elongation	2 Specimens	-	3.4.2	4.5.1.5
	Microstructure	1	-	3.4.1	4.5.7
	Forging	*		3.8.1	*
	Obturator (Dwg. 10542907)				
	Examination for Defects	5	-	3.1	4.4.2.2
	Adapter, Base (Dwg. 9352630)				
	Examination for Defects	5	-	3.1	4.4.2.7
	Yield Strength & Percent Elongation	2 Specimens	-	3.1	4.5.1.6

NOTES:

*See appropriate requirement paragraph.

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Table 1. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-P-64008 (AR)

PARAGRAPH	TITLE: CONTINUATION SHEET		SHEET 2 OF 3		DRAWING NUMBER: --- / NEXT HIGHER ASSEMBLY: ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Base (Dwg. 9352629)				
	Examination for Defects	5		3.1	4.4.2.6
	Yield Strength & Percent Elongation	2		3.4.2	4.5.1.5
	Protective finish test	1		3.1	4.5.10
	Microstructure	1		3.4.1	4.5.7
	Forging	*		3.8.1	*
	Metal defective (Prior to drilling wrench slots)	5		3.5.1	4.5.4
	Verification of heat treat (Prior to protective finish)	5		3.1	4.5.13
	Base and Adapter Assembly (Dwg. 9352631)				
	Examination for Defects	5		3.1	4.4.2.8
	Air leak test	5		3.1	4.5.9
	Protective finish test	1		3.1	4.10
	Body (Dwg. 9352632)				
	Examination for Defects	5	-	3.1	4.4.2.4
	Yield Strength & Percent Elongation	2	-	3.2.3	4.5.1.4

NOTES:

*See appropriate requirement paragraph.

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Table 1. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
-	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile Subassembly (Dwg. 93522633)				
	Examination for Defects	5	-	3.1	4.4.2.5
	Projectile Metal Parts (Dwg. 9352634)				
	Examination for Defects	25	-	3.1	4.4.2.9
	Salt Spray Test	2	-	3.1	4.5.10
	Proving Ground Function Test		-	3.7	4.5.11
	A. Excess pressure	10	-	3.7	4.5.11.1
	B. Zone 3	5	-	3.7	4.5.11.2
	C. Service pressure	5	-	3.7	4.5.11.3

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of projectiles shall contain:

a. Bodies from not more than one forging manufacturer.
b. Ogives from not more than one forging manufacturer.
c. Base and adapter assemblies from not more than one forging manufacturer of each component.

4.4.1.1 Heat treatment lot (body and base adapter). A heat treatment lot of bodies or base adapters shall consist of bodies or base adapters from one mill heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature and for the same time cycles in one unchanged process. The maximum lot size shall be 2000 bodies or 2000 base adapters. If a mill heat exceeds 2000, the mill heat shall be divided into any convenient number of heat treatment lots that would preclude the need for forming miscellaneous heat treatment lots. The heat treatment lot thus constituted shall be the basis for selecting samples for mechanical properties tests. Bodies or base adapters in each heat treatment lot shall be identified to that particular lot until satisfactory completion of mechanical properties tests.

4.4.1.2 Miscellaneous heat treatment lot (body and base adapter). Bodies or base adapters manufactured from small heats of steel, or (bodies/adapters)that have become separated from previously delivered lots may be grouped into a miscellaneous heat treatment lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.1. The maximum lot size shall be 2000 bodies or 2000 base adapters.

4.4.1.3 Heat treatment lot (base and ogive). A heat treatment lot of bases or ogives shall consist of bases or ogives manufactured from material of the same alloy and temper from one manufacturer. The material shall be heat treated by the same process, in the same equipment and using the same temperature and time cycles. At no time shall the lot quantity exceed 4800. The heat treatment lot thus constituted shall be the basis for selecting samples for mechanical properties tests. Components in each heat treatment lot shall remain segregated and identified until satisfactory completion of mechanical properties tests.

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4.4.1.4 Miscellaneous heat treatment lot (base and ogive). Ogives or bases manufactured from small lots of material or those that have become separated from previously heat treated lots may be grouped into a miscellaneous lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.3. The lot thus constituted shall be the basis for selecting samples for the mechanical properties tests.

4.4.1.5 Projectile lot. Heat treatment lots that have been determined to have complied with the yield strength and percent elongation requirements of the applicable drawing may be regrouped for assembly into projectile lots.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Automatic gaging. Facilities equipped with government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capability in accordance with the contract.

c. Tool control dimensions. Dimensions marked tool control shall be gaged during first article testing, then after eight hours production on each machine in operation and prior to and after any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor.

d. Process control dimensions. The contractor shall prepare a process control plan for the dimensions marked process control. The plan shall be submitted to the technical agency for approval prior to use. Dimensions to be gaged for process control are to be assured to a .40% AQL.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.1	Ogive				10542916
					NEXT HIGHER ASSEMBLY
					9352633
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.5	Visual/4.5.7
Major					
101	Pitch diameter of external thread, min.		0.40%	3.1	Gage
102	Major diameter of external thread, min.		0.40%	3.1	Gage
103	Diameter of cavity, rear		0.40%	3.1	Gage
104	Length from rear face to forward inside basic diameter		0.40%	3.1	Gage
105	Length from rear face rear inside basic diameter		0.40%	3.1	Gage
106	Variation in wall thickness		0.40%	3.1	Gage
107	Pitch diameter of fuze thread		0.40%	3.1	Gage
108	Minor diameters of fuze thread,max.		0.40%	3.1	Gage
109	Second diameter rear of fuze thread undercut		0.40%	3.1	Gage
110	First diameter rear of fuze thread undercut		0.40%	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.1	CONTINUATION SHEET				19542916
					NEXT HIGHER ASSEMBLY
					9352633
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111	Depth from front face to start of second diameter rear of fuze thread undercut		0.40%	3.1	Gage
112	Length from front face to forward basic O.D.		0.40%	3.1	Gage
113	Length from front face to rear basic O.D.		0.40%	3.1	Gage
114	Length of fuze thread, min.		0.40%	3.1	Gage
115	True position of second diameter rear of fuze thread undercut with pitch diameter of fuze thread		0.40%	3.1	Gage
116	True position at rear of fuze thread undercut with pitch diameter of fuze thread		0.40%	3.1	Gage
117	Depth from rear face to bottom of cavity		0.40%	3.1	Gage
118	Runout of outer ogive surface to body thread pitch diameter and shoulder (2 places)		0.40%	3.1	Gage
119	Runout of fuze thread pitch diameter projections (2 Places) to body thread pitch diameter, body shoulder and fuze mating surface		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.1	CONTINUATION SHEET				10542916
					NEXT HIGHER ASSEMBLY
					9352633
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
120	Yield strength and percent elongation	(a)	-	3.4.2	4.5.1.5
121	Microstructure	(a)	-	3.4.1	4.5.7
122	Protective finish	(b)	-	3.1	4.5.10
123	Hydrostatic test		100%	3.1	4.5.6
Minor					
201	Diameter of body thread undercut		0.65%	3.1	Gage
202	Length from front face to shoulder		0.65%	3.1	Gage
203	Depth from front face to rear of fuze thread undercut		0.65%	3.1	Gage
204	Diameter of fuze thread undercut		0.65%	3.1	Gage
205	Length, total		0.65%	3.1	Gage
206	Radii of chamfers missing or incorrect		0.65%	3.1	Visual
207	Surface finish improper		0.65%	3.1	Visual
208	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: (a) Sample size as specified in applicable test method paragraph.
(b) Sample size as specified in 4.4.3.3.1.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 10542907
4.4.2.2	Obturator				NEXT HIGHER ASSEMBLY 9352634
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of rear I.D. with forward I.D.		0.40%	3.1	Gage
102	Diameter outside, front		0.40%	3.1	Gage
103	Diameter outside, rear		0.40%	3.1	Gage
104	Diameter inside, front		0.40%	3.1	Gage
105	Depth of forward inside diameter		0.40%	3.1	Gage
106	Thickness of wall at rear		0.40%	3.1	Gage
107	Length, total		0.40%	3.1	Gage
Minor					
201	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9352632
4.4.2.3	Body (Prior to Banding)				NEXT HIGHER ASSEMBLY 9352633
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective in band seat area		100%	3.5.2	Visual
Major					
101	Diameter of rotating band seat		0.40%	3.1	Gage
102	Location of rotating band seat		0.40%	3.1	4.5.2
Minor					
201	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
202	Surface finish of band seat improper		0.65%	3.1	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 5		DRAWING NUMBER 9352632
4.4.2.4	Body				NEXT HIGHER ASSEMBLY 9352633
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective			3.5.1	4.5.3.2
Major					
101	Band seat metal defective	(a)		3.5.2	4.5.3.1
102	Hardness test of body		100%	3.2	4.5.1.2
103	Yield strength and percent elongation	(a)		3.2	4.5.1.4
104	Band chemical analysis	(a)		3.1	4.5.5
105	Runout of body undercut with bourrelets		0.40%	3.1	Gage
106	Runout of forward body ogive with bourrelets		0.40%	3.1	Gage
107	Runout of rear body ogive with bourrelets		0.40%	3.1	Gage
108	Runout of rotating band lip with bourrelets		0.40%	3.1	Gage
109	Runout of pitch diameter of base thread with bourrelets and rear face		0.40%	3.1	Gage
110	Runout of rear outside diameter with bourrelets		0.40%	3.1	Gage

NOTES:

(a) Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 5		DRAWING NUMBER 9352632
4.4.2.4	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY 9352633
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111	Runout of body cavity with bourrelets		0.40%	3.1	Gage
112	True position of body ogive thread counterbore with minor diameter of body ogive thread		0.40%	3.1	Gage
113	Pitch diameter of body ogive thread, min.		0.40%	3.1	Gage
114	Minor diameter of body ogive thread, max.		0.40%	3.1	Gage
115	O.D. of undercut		0.40%	3.1	Gage
116	Length to forward body ogive basic diameter		0.40%	3.1	Gage
117	Length to rear body ogive basic diameter		0.40%	3.1	Gage
118	Length total		0.40%	3.1	Gage
119	Length from rear face to front of rear bourrelet		0.40%	3.1	Gage
120	Length from rear face to rear of forward bourrelets		0.40%	3.1	Gage
121	Location of forward cannelure groove		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 5		DRAWING NUMBER 9352632
4.4.2.4	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY 9352633
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
122	Location of rear cannelure groove		0.40%	3.1	Gage
123	O.D. of bourrelets, min. (front & rear)		0.40%	3.1	Gage
124	O.D. of rotating band lip		0.40%	3.1	Gage
125	O.D. of rotating band flats		0.40%	3.1	Gage
126	O.D. of rotating band cannelures		0.40%	3.1	Gage
127	O.D. of obturator seat		0.40%	3.1	Gage
128	I.D. of body cavity		0.40%	3.1	Gage
129	O.D. rear of body		0.40%	3.1	Gage
130	Depth of body cavity keyway		0.40%	3.1	Gage
131	Location to bottom of body cavity keyway		0.40%	3.1	Gage
132	Length of body cavity		0.40%	3.1	Gage
133	Length of body cavity keyway from rear face		0.40%	3.1	Gage
134	Length from rear face to rear of rotating band		0.40%	3.1	Gage
135	Threads damaged or not full		0.40%	3.1	Gage
136	Surface straightness of rear face of projectile		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	CONTINUATION SHEET	4 OF 5	9352632
			NEXT HIGHER ASSEMBLY 9352633

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
137	Thickness of wall at forward end of cavity		0.40%	3.1	Gage
138	Width of rotating band		0.40%	3.1	Gage
139	Width of rotating band relief groove		0.40%	3.1	Gage
140	Width of body cavity keyway		0.40%	3.1	Gage
141	Band seat visible in rotating band cannelures		0.40%	3.3.2	Visual
142	Pitch diameter of base thread		0.40%	3.1	Gage
143	Minor diameter of base thread,max.		0.40%	3.1	Gage
144	True position of base thread counterbore with pitch diameter of base thread		0.40%	3.1	Gage
145	Length of base thread, min.		0.40%	3.1	Gage
146	Length from front of band to band basic diameter		0.40%	3.1	Gage
147	Runout of pitch diameter of ogive & thread to bourrelets and ogive mating surface		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-P-64008 (AR)

PARAGRAPH	TITLE		SHEET 5 OF 5		DRAWING NUMBER
4.4.2.4	CONTINUATION SHEET				9352632
					NEXT HIGHER ASSEMBLY
					9352633
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Diameter of body ogive thread counterbore		0.65%	3.1	Gage
202	Depth of body ogive thread counterbore		0.65%	3.1	Gage
203	Diameter of base thread counterbore		0.65%	3.1	Gage
204	Depth of base thread counterbore		0.65%	3.1	Gage
205	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
206	Presence and size of undercut forward of rotating band (if applicable)		0.65%	3.1	Visual-Gage
207	Surface finish improper		0.65%	3.1	Visual
208	Protective coating missing, incomplete or damaged		0.65%	3.1	Visual
209	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9352633
4.4.2.5	Projectile, Subassembly				NEXT HIGHER ASSEMBLY 9352634
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective stamp missing (indicating failure to pass test)		100%	3.1	Visual
2	Metal defective		100%	3.5	Visual (probe light)
Major					
101	Weight		0.40%	3.1	Approval Scale
102	Runout of pitch diameter of fuze thread with bourrelets and front face		0.40%	3.1	Gage
103	Runout forward of ogive with bourrelets		0.40%	3.1	Gage
104	Runout rear of ogive with bourrelets		0.40%	3.1	Gage
105	Gap between ogive and body		0.40%	3.1	Gage
106	Length total, max.		0.40%	3.1	Gage
107	Air leak test		100%	3.1	Gage
108	Location of cannister locating hole		0.40%	3.1	Gage
109	Diameter of cannister locating hole		0.40%	3.1	Gage
110	Depth of cannister locating hole		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.6	Base				9352629
					NEXT HIGHER ASSEMBLY 9352631
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective (Prior to machining wrench slots)		100%	3.5	4.5.4
Major					
101	True position of "O" ring groove diameter with pitch diameter of thread		0.40%	3.1	Gage
102	True position of diameter forward and rear of "O" ring groove with pitch diameter of thread		0.40%	3.1	Gage
103	Runout of external taper with pitch diameter of threads (2 locations)		0.40%	3.1	Gage
104	Length to basic diameter on external taper from underside of shoulder		0.40%	3.1	Gage
105	Depth from rear face to basic diameter on cavity sidewall		0.40%	3.1	Gage
106	Length from front face (thread face) to underside of shoulder		0.40%	3.1	Gage
107	Distance between front face and bottom of cavity in rear face		0.40%	3.1	Gage

NOTES:

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PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 9352629
4.4.2.6	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY 9352631
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
108	Pitch diameter of thread, min.		0.40%	3.1	Gage
109	Pitch diameter of thread, max.		0.40%	3.1	Gage
110	Major diameter of thread, min.		0.40%	3.1	Gage
111	Diameter of "O" ring groove		0.40%	3.1	Gage
112	Diameter forward and rear of "O" ring groove		0.40%	3.1	Gage
113	Width of "O" ring groove		0.40%	3.1	Gage
114	Diameter of cavity in front face		0.40%	3.1	Gage
115	Depth of cavity in front face		0.40%	3.1	Gage
116	Length from front of lip to rear of "O" ring groove		0.40%	3.1	Gage
117	Length from rear of "O" ring groove to underside of shoulder		0.40%	3.1	Gage
118	Length from rear face to underside of shoulder		0.40%	3.1	Gage
119	True position of slot(s) diameter with pitch diameter of thread		0.40%	3.1	Gage
120	Surface straightness at underside of shoulder		0.40%	3.1	Gage
121	Threads damaged		0.40%	3.1	Gage
122	Wrench slot missing (2)		0.40%	3.1	Visual
123	Yield strength & percent elongation	(a)		3.4.2	4.5.1.5

NOTES:
(a) Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
4.4.2.6	CONTINUATION SHEET		9352629
			NEXT HIGHER ASSEMBLY 9352631

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
124	Protective finish	(b)		3.1	4.5.10
125	Verification of heat treat		100%	3.1	4.4.3.2/4.5.1.3
126	Microstructure	(a)		3.4.1	4.5.7
Minor					
201	Diameter of thread undercut		0.65%	3.1	Gage
202	Width of thread undercut, max.		0.65%	3.1	Gage
203	Length from rear face to bottom of wrench slot		0.65%	3.1	Gage
204	Diameter of wrench slots (2)		0.65%	3.1	Gage
205	Distance between wrench slots		0.65%	3.1	Gage
206	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
207	Surface finish improper		0.65%	3.1	Visual
208	Protective coating damaged or missing		0.65%	3.2	Visual
209	Radii or O-ring groove	(c)		3.1	Gage
210	Poor workmanship			3.10	Visual

NOTES: (a) Sample size as specified in applicable test method paragraph.
(b) Sample size shall be as specified in 4.4.3.2.1 herein.
(c) See paragraph 4.4.2.c herein.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 9352630
4.4.2.7	Adapter, Base				NEXT HIGHER ASSEMBLY 9352631
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Runout of outside taper with pitch diameter of external threads and front face of flange		0.40%	3.1	Gage
102	Runout of internal thread pitch diameter projections (2) with external thread pitch diameter and shoulder and rear face		0.40%	3.1	Gage
103	Full thread (external), min.		0.40%	3.1	Gage
104	True position of "O" ring groove diameter with pitch diameter of external threads		0.40%	3.1	Gage
105	Pitch diameter of external thread,min.		0.40%	3.1	Gage
106	Pitch diameter of external thread,max.		0.40%	3.1	Gage
107	Major diameter of external thread,min.		0.40%	3.1	Gage
108	Pitch diameter of internal thread,max.		0.40%	3.1	Gage
109	Pitch diameter of internal thread,min.		0.40%	3.1	Gage
110	Minor diameter of internal thread,max.		0.40%	3.1	Gage
111	Diameter of "O" ring groove rear of external thread		0.40%	3.1	Gage
112	Width of "O" ring groove rear of external thread		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 9352630
4.4.2.7	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY 9352631
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	Length from front face to front face of flange		0.40%	3.1	Gage
114	Length from rear of external thread to front face of flange		0.40%	3.1	Gage
115	Width of flange		0.40%	3.1	Gage
116	Diameter of counterbore rear of internal thread		0.40%	3.1	Gage
117	Length from rear face to bottom of counterbore		0.40%	3.1	Gage
118	Length from rear face to top of counterbore		0.40%	3.1	Gage
119	Runout of counterbore with pitch diameter of external thread and front face of flange		0.40%	3.1	Gage
120	Surface straightness at forward face of flange		0.40%	3.1	Gage
121	Surface straightness at rear face		0.40%	3.1	Gage
122	Yield strength & percent elongation	(a)		3.1	4.5.1.6
123	Protective finish missing		0.40%	3.1	Visual

NOTES:
(a) Sample size as specified in applicable test method paragraph.

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	CONTINUATION SHEET	3 OF 3	9352630
			NEXT HIGHER ASSEMBLY 9352631

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Length from forward face of flange to basic		0.65%	3.1	Gage
202	Diameter of portion behind external thread		0.65%	3.1	Gage
203	Diameter on taper		0.65%	3.1	Gage
204	Radii or chamfer missing or incorrect		0.65%	3.1	Gage
205	Surface finish improper		0.65%	3.1	Gage
206	Radii of O-ring groove	(b)		3.1	Gage
207	Poor workmanship		0.65%	3.10	Visual

NOTES:
(b) See paragraph 4.4.2.c herein.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.8	Base and Adapter Assembly				9352631
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Surface acceptance stamp missing from base			3.1	Visual
Major					
101	Air pressure leak test		100%	3.1	4.5.9
102	Protective finish	(a)	-	3.1	4.5.10
103	Gap between base and adapter, max.		0.40%	3.1	Gage
104	Location of pins (2)		0.40%	3.1	Gage
105	"O" ring missing (Prior to assembly of adapter)		0.40%	3.1	Visual
106	Pins above flush		0.40%	3.1	Visual
107	Either pin missing		0.40%	3.1	Visual
108	External adapter thread damaged		0.40%	3.1	Visual
109	Heat treat stamp missing from base		100%	3.1	Visual
110	Torque of adapter to base improper		0.40%	3.1	Gage

NOTES:

(a) Sample size shall be as specified in 4.4.3.3.2 herein.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.8	CONTINUATION SHEET				9326331
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Paint on threads or prohibited surface		0.65%	3.1	Visual
202	Paint missing		0.65%	3.1	Visual
203	Thread sealing compound missing (Prior to assembly)		0.65%	3.1	Visual
204	Protector cap missing		0.65%	3.1	Visual
205	Grease missing from O-ring		0.65%	3.1	Visual
206	Poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.9	Projectile, Metal Parts Assembly				9352634
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Proving Ground Test				
	a. Loss or break up of metal part in flight	(a)		3.7	4.5.11
Major					
101	Protective finish	(b)		3.1	4.5.10
102	Diameter of bourrelets, max.		100%	3.1	Gage
103	Rotating band damaged		0.40%	3.1	Visual
104	Obturator missing or damaged		0.40%	3.1	Visual
105	Protective finish incomplete or damaged		0.40%	3.1	Visual
106	Proving Ground Test				
	a. Base or cargo not ejecting	(a)		3.7	4.5.11
Minor					
201	Paint on rotating band bearing surfaces in excess of drawing allowances		0.65%	3.1	Visual
202	Paint on threads		0.65%	3.1	Visual
203	Paint on obturator		0.65%	3.1	Visual
204	Paint not dry		0.65%	3.1	Visual/Manual
205	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES: (a) Sample size as specified in 4.4.3.4.
(b) Sample size as specified in 4.4.3.3.3.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.9	CONTINUATION SHEET				9352634
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
206	Grommet missing or loose (See Note 2)		0.65%	3.1	Visual/Manual
207	VCI paper missing (See Note 2)		0.65%	3.1	Visual/Manual
208	Protective caps (Base & Fuze Thds) missing		0.65%	3.1	Visual/Manual
209	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

Note 2: Not required for intraplant use.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Pallet*				8837839
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Pallet damaged (inserviceable)		0.65%	3.1	Visual
202	Improperly packed		0.65%	3.1	Visual
203	Strapping missing, loose or broken		0.65%	3.1	Visual/Manual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual
205	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

*Not required for intraplant use.

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4.4.3 Testing. Testing is described in the First Article Table, Quality Conformance Inspection Tables, and herein.

4.4.3.1 Hardness prior to cold forming (body). When the contractor elects to cold form the body, he will assure compliance with paragraph 3.2.1. The hardness test shall be conducted at a midpoint along the length of outside diameter of the body. Based upon conformance with the requirement and after 20,000 consecutive bodies have been subjected to this test, the contractor may propose an alternate hardness sampling procedure. As a minimum, the alternate hardness sampling procedure shall include hardness testing of the first twenty consecutive bodies out of a slow cool after each start up and sampling for hardness on a specific hourly basis. The alternate sampling procedure shall be submitted to the procuring activity for approval together with hardness data for approval.

4.4.3.2 Verification of aluminum base heat treat. After heat treat and prior to protective coating, each base shall be verified for its having been heat treated. Bases failing to comply shall be rejected. Rejected parts may be heat treated and resubmitted. All bases passing this test shall be stamped in accordance with the applicable drawing. Testing shall be in accordance with paragraph 4.5.1.3.

4.4.3.3 Protective finish

4.4.3.3.1 Base and ogive. One base and one ogive for each twenty four hours of continuous production or fraction thereof shall be subjected to salt spray test as required by the applicable drawing. If failure occurs, all bases or ogives processed since the last acceptance shall be rejected. The test shall be performed in accordance with 4.5.10.

4.4.3.3.2 Base and adapter assembly. Two assemblies from each week's production shall be subjected to salt spray test to determine compliance with the applicable drawing. One assembly shall be selected from the first thirty minutes of production. The second sample may be taken at any time, but not less than two days after the first sample is taken. If failure occurs, all items processed since the last acceptance shall be rejected. The test shall be performed as specified in 4.5.10.

4.4.3.3.3 Projectile, 155MM, XM825, metal parts assembly (Prior to assembling obturator and grommet). Two samples per week shall be subjected to salt spray test in accordance with 4.5.10. The first sample shall be taken from the first 30 minutes of production. The second sample may be taken at any time, but not less than two days after the first sample is taken. If a failure occurs, all items processed since the last acceptance shall be rejected.

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4.4.3.4 Proving ground test. Eight assemblies from each lot shall be selected for this test. Three of eight assemblies shall be tested at excess pressure, two at Zone 3, and three at service pressure. If any assembly fails to comply with all applicable requirements, the lot represented by that assembly shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is identified in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1 and the test method paragraph (see 4.5). The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4.4.1 Automatic inspection equipment. All automatic inspection equipment, when utilized, shall be subjected to periodic verification and calibration check during production. The calibration procedures shall be documented and have prior written approval of the technical agency. Procedures shall include as a minimum: Frequency of calibration, method of calibration, and the method of retrieval of items produced between calibration. These procedures shall be included in the operation and or calibration procedures required by 4.4.4.

4.4.4.2 Certification of nondestructive test personnel. All personnel operating or calibrating nondestructive test equipment shall be qualified and certified in accordance with MIL-STD-410.

4.5 Methods of inspection.

4.5.1 Mechanical properties.

4.5.1.1 Hardness prior to cold forming (body). The surface of each body in the area to be tested shall be prepared by machining or grinding a flat or sufficient depth, .030 inches minimum, to remove scale and get below the decarburization zone. Test shall be performed using the procedure of ASTM Method E-103. Bodies failing to comply with 3.2.1 may be re-annealed and retested.

4.5.1.2 Hardness (body). In order to determine uniformity of heat treatment, each body after application of the welded overlay band and heat treatment and before final machining shall be subjected to a hardness test with a Brinell Hardness Tester or any other method approved by the procuring activity. Readings shall be taken on the body approximately three-quarter inch forward of the rotating band. The test shall be performed in accordance with ASTM Method E-103 and the following:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding a flat of sufficient depth, .015 inches minimum, to remove scale and get below the decarburization zone. Flats and impressions shall be completely removed by final machining.

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b. Hardness tester periodic verification shall be performed using Brinell test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically recorded.

d. The hardness tester shall use a standard 10mm Brinell ball. The hardness tester shall determine hardness in accordance with the following test sequence. First, the tester shall clamp the body to be tested with a 10,000-500 lb. clamping load. Second, apply a minor load of approximately 10% of the 3000 kg. of total test load. Third, apply the total test load. The clamping load will not be exceeded when the test load is applied. Fourth, relax the total test load back to the minor load. Fifth, automatically measure this penetrator travel from initial minor load position to new minor load position following application of the total test load. Sixth, display hardness and release the body.

4.5.1.3 Verification of heat treat of base. Testing shall be with eddy current or any other approved method.

4.5.1.4 Yield strength and percent elongation of body. The hardest and two (2) softest bodies from each standard heat treatment lot or the two (2) hardest and four (4) softest bodies from each miscellaneous heat treatment lot as determined by the hardness test in 4.5.1.2 shall be subjected to this test. Sample for mechanical property testing shall be selected after heat treatment of the entire heat treatment lot. Two tension test specimens conforming to the largest obtainable round type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of each body, midwall of the area of the body one inch from the front end. The number of specimens and their location are for the convenience of the contractor and government and shall not restrict the contractor or government from testing specimens at other locations to determine that the mechanical properties of the body meet the drawing requirements. The criteria for performing the tests shall be as prescribed by ASTM Method E-8. Failure of any test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the contractor, the entire lot may be selected and tested from a segment of the heat treat lot having a tighter hardness range than the heat treatment lot as originally sampled. If an acceptable hardness range can be established for this heat treatment lot, bodies falling within this range will be accepted. All other bodies will be rejected. Rejected bodies may be reheat treated and retested.

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4.5.1.5 Yield strength and percent elongation of ogive and base. The sample size for the tension test shall be one component from each 1200 parts or portion thereof from each heat treatment lot. Two longitudinal test specimens conforming to the largest obtainable round specimen as prescribed by ASTM Method E-8 shall be taken from the sidewall of each sample. The base specimens shall be taken from midwall centered between rear and front ends 180° apart. The ogive tension test specimen shall be taken from the midwall at a location where the gage length center of the specimen is approximately 1 inch from the finished rear face of the body thread end of the ogive, 180° apart. The criteria for performing the tests shall be as prescribed by ASTM Method E-8. Percent elongation shall be recorded to the nearest tenth percent. Failure of any specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the quantity represented by the sample.

4.5.1.6 Yield strength and percent elongation of base adapter. A section of tubing after heat treatment from each heat treatment lot shall be subject to this test. Two (2) tension test specimens shall be taken longitudinally from opposite sides of the tubing. The number of specimens and their location are for the convenience of the contractor and government and shall not restrict the contractor or government from testing specimens at other locations to determine that the mechanical properties of the base adapter meets the drawing requirements. The criteria for performing the test shall be as prescribed by ASTM Method E-8. The percent elongation shall be recorded to the nearest tenth percent. Failure of any specimen to comply with the yield strength or elongation requirements in the applicable drawing shall be cause for rejection of the lot.

4.5.2 Rotating band seat location. The contractor may use any control method that will assure the band seat meets the dimensional requirements of the applicable drawing. The inspection plan must include all the required criteria to assure the dimensional aspects of the drawing will be equal tc the characteristics defined in 4.4.2.3. When the contractor establishes a control method, the manufacturing procedure, and inspection plan shall be forwarded to the Technical Agency for approval prior to use. No change to these controls will be made once established without approval from the Technical Agency.

4.5.3 Inspection for body metal defects after machining.

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4.5.3.1 Magnetic particle inspection of band seat. The first five banded bodies, after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to this test to comply with the visual standards of 3.5.2. Reducing sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been visually examined and found acceptable. Reduced sampling of one in each 8 hours production or fraction thereof per welding machine in operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been visually examined and found acceptable (see 6.9).

The samples provided shall have the band removed by machining the band and chemically etching without damaging the base metal (steel) of the band seat (see 6.7). Each body sampled shall then be inspected using the magnetic particle inspection technique.

The procedure for performing this test shall be as specified in MIL-I-6868 for wet fluorescent method with an indirect magnetization by either the continuous or residual method, inducing a circular magnetic field, and again after inducing a longitudinal magnetic field.

Should two consecutive failures occur in any welding machine(s), in any reduced sample size, the machine(s) shall revert back to the original first five consecutive banded bodies per that welding machine(s) and after five consecutive acceptable bodies, the sequence of reduced sampling will also apply. Banded bodies shall be identified along with the welding machine that applied the band.

4.5.3.2 Finished body. Bodies after final machining shall be inspected by the magnetic particle inspection technique. The procedures for performing this test shall be as specified in MIL-I-6868, for wet fluorescent method with indirect magnetization by either the continuous or residual methods. The entire body, inside and outside, shall be inspected after inducing a circular magnetic field and again after inducing a longitudinal magnetic field. The magnetic particle standard per dwg. 9289318 shall be utilized to establish inspection sensitivity. Any body which fails to comply with the requirements of 3.5 shall be rejected. Bodies which comply shall be stamped with the letter "M" as specified on the applicable drawing.

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4.5.3.3 Magnetic particle suspension. The solid content of the suspension shall be maintained at the level recommended by the suspensoid manufacturer. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by Section 4 of MIL-I-6868. When performing this test for solid content of the suspension, if it is determined that it does not comply, all bodies that have been tested and accepted per 4.5.3.2 since the last satisfactory test was performed shall be retested after the suspension has been brought to the proper level. The magnetic particle test shall be performed on equipment that complies with the requirements of specification MIL-M-6867. The operator(s) of the equipment and the inspector(s) making the determinations shall be qualified and certified in accordance with MIL-STD-410.

4.5.4 Metal defective test (base). The base shall be inspected for metal defects using the eddy current method. Bases failing to comply with the requirements of 3.3.5 shall be classed defective and removed from the lot. All bases passing this test shall be stamped in accordance with ther applicable drawing. The standards and test method shall be in accordance with drawings 9280354 and 9280355.

4.5.5 Band chemical analysis. The first five banded bodies, after the welded overlay band has been applied, for each welding machine in opeation, shall be subjected to this test. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation has been tested and found acceptable. Reduced sampling of one in each 4 hours production or fraction thereof per welding machine in operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been tested and found acceptable. Sample chips shall be obtained from each band from the last .050 inches removed when machining to the finished diameter. All chips shall be collected and identified with the shell from which the chips were machined. A minimum 5 gram sample from each complete specimen thus obtained from each band shall be dissolved and the analysis made from the individual solutions. The sample may not be cleaned with a magnet. If failure occurs sampling will revert to the original first five consecutive banded bodies per that machine(s) and after five consecutive acceptable bodies, the sequence of reduced sampling will also apply. Failure to comply with the requirements of the drawing shall be cause for rejection of the production represented by the sample.

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4.5.6 Hydrostatic test of ogive. Prior to the application of protective finish, the finish machined ogive shall be subjected to a hydrostatic test. The ogive shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure is reached. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours of continuous operation. It shall also be calibrated prior to re-starting the operation after a layoff of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the test pressure was actually lower than that specified on the applicable drawing, the ogives tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. The fixture should be such that it does not restrain the ogive. That is, it shall simulate actual assembly conditions.

4.5.6.1 Lot sampling plan. 100% hydrostatic testing of finish machined ogives shall be conducted until 1200 ogives consecutively pass the hydrostatic test. Subsequently lots of no greater than 5000 ogives may be submitted for testing. A sample of 50 ogives per lot shall then be tested. Failure of any ogive to meet the requirement of the applicable drawing shall be cause for rejection of the lot represented by the sample. When a lot has been rejected while on sampling inspection, the frequency shall revert back to the 100 percent inspection until 1200 ogives consecutively pass the hydrostatic test, to qualify for reduced sampling testing.

4.5.6.2 Continuous sampling (option). The ogive hydrostatic testing may be conducted using a continuous sampling plan from MIL-STD-1235, CSP-1 with AQL of 0.10.

4.5.7 Microstructure examination (ogive & base). One ogive and one base forging from each heat treatment lot shall be selected for microstructure examination. The metallographic specimen shall be etched with Keller's Reagent and all areas examined under a microscope/metal-lograph at a minimum magnification of 250 diameters. Any evidence of the insoluble phase generally occurring in a semi-continuous or continuous network form shall be cause for rejection of the lot represented by the sample.

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Two examples of acceptable microstructures are shown in Figures 1a and 1b. In Figure 1a, the insoluble phase is well broken-up randomly distributed within a matrix which is essentially unrecrystallized. In Figure 1b, the insoluble phase is well broken-up and randomly distributed within a matrix which is partially recrystallized. Two examples of the unacceptabld microstructures are shown in Figures 2a and 2b. In Figure 2a, the insoluble phase is distributed in a semi-continuous form within the matrix which shows microsegregation (or coring). In Figure 2a, the insoluble phase is distributed in a more continuous form within the matrix which shows microsegregation (or coring). Either of the above structures (Figures 2a and 2b) must be avoided because they are detrimental to the integrity of the material.

Microstructural preparation and examination shall be performed by a metallographer or metallurgical engineer familiar with aluminum alloys.

4.5.7.1 Ogive specimen. A metallographic specimen at least 1/2 inch in length shall be removed from the sidewall. It shall be a longitudinal cross section of the body thread area with the center of the specimen approximately 3/4 inches from the finished rear face.

4.5.7.2 Base specimen. A metallographic specimen at least 1/2 inch in length shall be removed from the base and be a longitudinal cross section with its center located at mid-diameter of the base.

4.5.8 Air leak test (projectile, subassembly). The ogive to body joint of the subassembly shall be tested for compliance with the requirement specified on the drawing. A force shall not be applied in the axial direction such as to compress the ogive to body joint. Failure to comply shall be cause to reject the subassembly. Subassemblies which comply shall be stamped with the letter A in 1/8 inch letters in the area indicated on the drawing. The test may be accomplished by one of the following procedures.

4.5.8.1 Submersion test (alternate 1). Subassemblies shall be mounted in an approved fixture in such a manner that the subassembly shall be submerged in water. The required air pressure shall be applied and held for the minimum specified time, after the water in the tank is permitted to settle after immersion. Observation for leakage shall be made for the entire time the subassembly is required to be pressurized. The presence of air bubbles in the vicinity of the body to ogive joint or rising to the surface of the water shall be cause for rejection of the subassembly.

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4.5.8.2 Pressure drop test (alternate 2). Each subassembly shall be mounted so as to seal the projectile above and below the ogive to body joint with a collar (5 in^3 max.) and pressurizing this collar at 15 psi for 15 seconds while looking for a maximum pressure drop of 0.01 psi during the 15 second hold cycle. Pressure source shall be inactive during the hold cycle. A pressure drop in excess of 0.01 psi shall be cause for rejection of the subassembly.

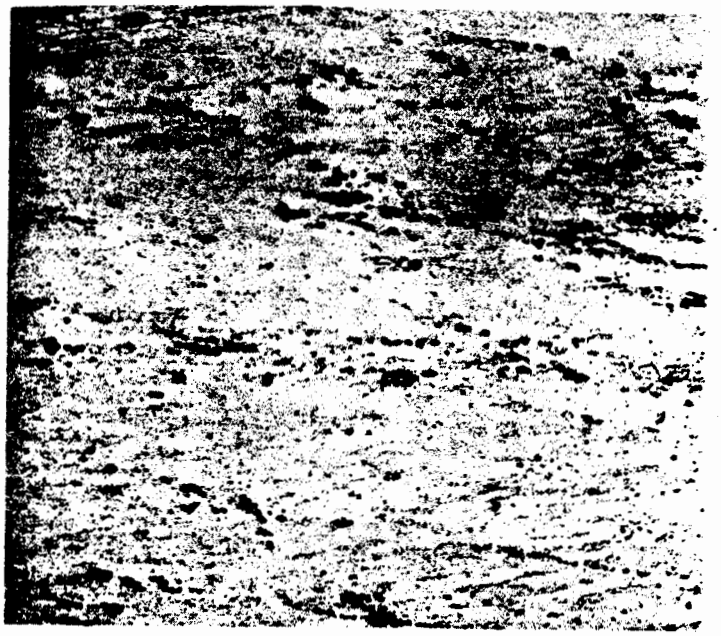
4.5.9 Air leak test (base and adapter assembly). The base to adapter joint of the base and adapter assembly shall be tested for compliance with the requirement specified on the drawing. The test fixture utilized shall not apply a force in the axial direction such as to compress the base to adapter joint. Any assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test may be accomplished by one of the following procedures:

4.5.9.1 Submersion test The base and adapter assembly shall be mounted in a fixture in such a manner that the assembly will be submerged in water. The required air pressure shall be applied and held for the minimum specified time after the water in the tank is permitted to settle after immersion in water. Observation for leakage shall be made for the entire time the base and adapter assembly is required to be pressurized. The presence of air bubbles in the vicinity of the base to adapter joint or rising to the surface of the water shall be cause for rejection.

4.5.9.2 Pressure drop test (alternate 2). Each base and adapter assembly shall be mounted so as to seal the assembly above and below the base to adapter joint with a collar (5 in^3 max.) and pressuring this collar at 15 psi for 15 seconds while looking for a maximum pressure drop of 0.01 psi during the 15 second hold cycle. A pressure drop in excess of 0.01 psi shall be cause for rejection of the assembly.

4.5.10 Protective finish test. Salt spray test shall be performed on the base, ogive, base and adapter assembly, and projectile subassembly utilizing the equipment, methods and procedures specified in ASTM B117 and in accordance with the applicable detailed specification for the protective finish except the curing cycle for the projectile subassembly shall be in accordance with dwg. 9352633.

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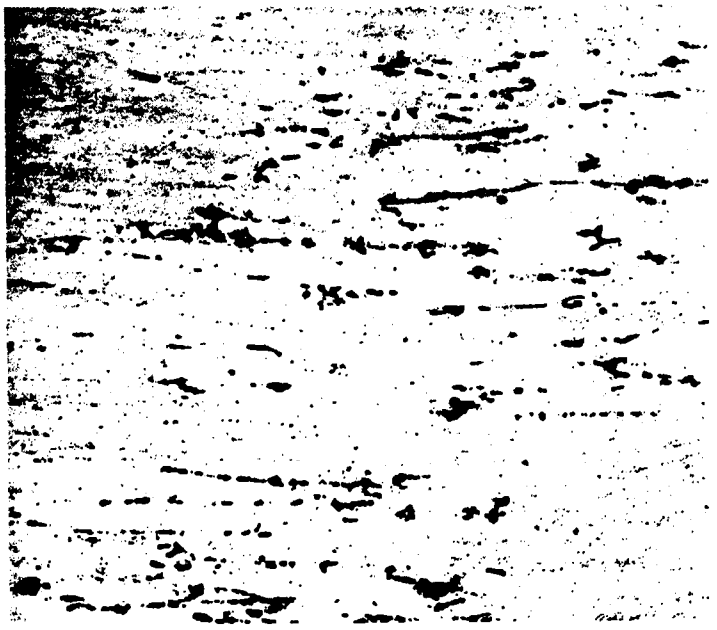
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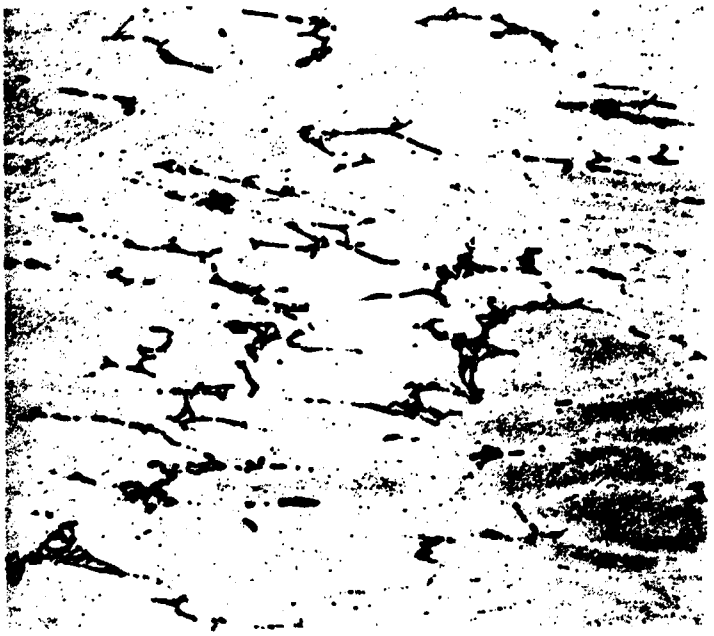
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FIGURE 1 Acceptable Microstructures (250x)

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a



b

FIGURE 2 Unacceptable Microstructures (250x)

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4.5.11 Proving ground ballistic tests. The following tests shall be conducted using Cannon tubes which have not reached the fourth quarter of remaining life based on current gun tube condemnation criteria. Photographic equipment shall be positioned near the muzzle of the weapon to record the condition of the projectile as it exits the tube. Chamber pressures and velocities shall be recorded for each round fired. All photographs and recovered hardware shall be examined for any noncompliance with the applicable requirements in 3.6. Range, deflection, and velocity shall be recorded for each round fired for information. An attempt shall be made to recover the projectiles fired for this test.

4.5.11.1 Excess pressure. Each projectile submitted for this test shall contain explosively inert WP/FELT wedge filled canister, have its base assembled, a dummy explusion charge, and a dummy fuze. Each projectile will be fired with a PXR-697 Proof Charge conditioned to 145°F to obtain a 53 to 57 thousand PSI chamber pressure with an M199 cannon tube.

4.5.11.2 Zone 3. Each projectile submitted for this test shall contain explosively inert WP/FELT wedge filled canister, have its base assembled, contain a live explusion charge, and an M577 fuze. Each projectile shall be fired in either an M185 of M199 Cannon with an M3A1 Propelling Charge. The projectile complete with explusion charge, shall be conditioned -40 ± 5°F. Fuze function time shall be recorded.

4.5.11.3 Service pressure. Each projectile submitted for this test shall contain explosively inert WP/FELT wedge filled canister, have its base assembled, a live explusion charge and an M577 fuze. Each projectile shall be fired in an M199 cannon tube with an M203 propelling charge conditioned at 70°F. Fuze function time will be recorded.

5. PACKAGING

5.1 Packing. This section (Section 5) is not applicable if LAP operations are to be performed at the same facility.

5.1.1 Level A. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Minimum Military Pack.

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5.2 Packaging and Packing.

5.2.1 Projectile metal parts assemblies shall be palletized in accordance with Drawing 8837839.

5.2.2 Bases shall be packaged and packed in accordance with MIL-STD-1169, and will conform to contractor drawings.

5.3 Marking. Marking of the pallet shall comply with Dwg. 8837839. Marking of the cartons for the bases shall comply with MIL-STD-1169 and shall include marking to assure that the base is not shipped or stored with the cavity facing upward. Bases need not be identified with projectile lot numbers.

6. NOTES

6.1 Intended use. The metal parts covered by this specification are intended for the Projectile, 155MM: XM825.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.4 Data requirements. Data Item Descriptions required for submission under this specification are as follows:

Paragraph	Data Requirement	Applicable DID
3.2.1, 3.2.3, 3.3.1, 3.4.3, 3.8.1, 4.5.2, 6.12	Procedure Report	* DI-T-5372 (Tailored)
4.5.1.1, 4.5.1.2, 4.5.1.3, 4.5.1.4, 4.5.1.5, 4.5.1.6, 4.5.7	Quality Inspection Test Demonstration & Evaluation Report	DI-R-1724
4.5.3.1, 4.5.3.2	Quality Inspection Report	DI-R-1721

* Delete a, b, c, d, or paragraph 1 block 10, Form 1664, and substitute contractor designated parameters and limits.

The above data shall be forwarded in accordance with the requirements of the contract to Commander, ARRADCOM, ATTN: DRDAR-QAR-Q, Dover, NJ 07801.

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6.5 Submission of ballistic test data and data cards. In addition to the normal distribution, two copies of all ballistic data and ammunition cards shall be forwarded to Commander, ARRADCOM, ATTN: DRDAR-QAR-Q, Dover, NJ 07801 and Commander, ARRADCOM, ATTN: DRCPM-CWA, Dover, NJ 07801.

6.6 Cold forming. The contractor is cautioned that if cold forming is utilized in processing of the XM825 body, those bodies which do not completely clean up in the machining operation prior to the cold forming have a greater propensity to develop circumferential tears during the forming operation particularly if areas of non-cleanup are heavily scaled.

6.7 Suggested etching procedures for removing gilding metal overlay.

a. Recommended. Immerse body in commercial copper stripper which is cyanide and chromate free and prepared in accordance with manufacturer's directions. In a fresh solution, an overlay layer 0.010 to 0.015 inch thick will dissolve in 10 to 14 hours. Renew solution as necessary.

b. Solution. 500 grams Chromic oxide $Cr\ 0_3$ (commercial flakes) 50 ml. sulphuric acid H_2SO_4 (technical grade) 950 ml. water H_2O.

This solution may be used by passing current through the solution (50 amperes dc) with the immersed shell as the anode and the container as the cathode. An overlay layer 0.010 to 0.015 inch thick will require approximately two to three hours to dissolve in a fresh solution. The solution can also be used without an external current source by immersing the shell for approximately 8 to 12 hours. An active solution would clean all pits.

6.8 The specified alloys, 1340 and 4140, which are compatible with the welded overlay rotating band, have been successfully used as body material. Experience to date has indicated that 1340 steel requires closer control over the welding process and thermal treatment to prevent cracks after banding. However, this does not preclude the need for similar close control over the welding process and thermal treatment with 4140 steel to be compatible with the production process.

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6.9 Recommended welding operation of overlay band be guided by Frankford Arsenal "Process Recommendations for Banding 155MM, M483 Projectile" dated 19 April 1973. Requests for copies should be submitted to: Commander, ARRADCOM, ATTN: DRDAR-LCU-M, Dover, NJ 07801.

6.10 Steel blooms, billets, slabs and bars

6.10.1 It is the responsibility of the manufacturer of the projectile metal parts to select a steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of the forger and machiner to produce the projectile to the dimensions, mechanical properties and metal integrity specified as an end item requirement.

6.10.2 The supplier of the selected steel raw material is responsible only for furnishing a metal composition, grade, and type with dimensions, weight, soundness and freedom from metallurgial or mechanical defect limitations identifiable with the standard AISI grade (selected by the projectile manufacturer) and complying with the specific ASTM designation and tests selected by the projectile manufacturer.

6.10.3 In the application of carbon steel, blooms, billets, slabs and bars, forging quality processing techniques adapted to use of the selected steel should be applied with the expectation that there will be variables in the metal cross-sectional dimensions, in weight per unit length, in the hardenabiity variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment in terms of equipment sequence of operations, tooling designs, handling devices, and processing techniques having the capabilities of compensating for such variables.

6.11 Adequate working of ogive material. In order to insure adequate working of the ogive material and the break-up of insolubles from the cast ingot to the final raw forging configuration, it is recommended that a minimum extrusion ratio of six (6) be used representing an 83% reduction of area from the cross-sectional area of the extrusion.

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6.12 Heat treatment procedure - Contractor shall submit, for information purposes, the heat treatment procedure per DI-T-5372.

6.13 To minimize the need for reheat treating material or for having miscellaneous heat treatment lots it is recommended that the contractor process a preproduction sample to establish what should be an acceptable hardness range for each mill heat. This hardness range could then be used for screening bodies during processing of the heat treatment lot from that mill heat. When using a hardness range, acceptance will still be in accordance with para. 4.5.1.4.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1320-A669)

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INCH-POUND

MIL-P-64008 (AR)
AMENDMENT 5
02 August 1993
SUPERSEDING
AMENDMENT 4
11 April 1991.

MILITARY SPECIFICATION

PROJECTILE, 155MM: M825 METAL PARTS ASSEMBLY

This amendment forms a part of Military Specification MIL-P-64008 (AR), dated 27 December 1982, and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.: SCOPE: Delete "cope" and substitute "SCOPE".

1.1: Delete "XM825" and substitute "M825/M825A1".

2.1.1: Add Military Specification:

"MIL-A-8625 - Anodic coatings, for Aluminum alloys"

2.1.1: Add Military Specification:

"MIL-A-70625 - Automated Acceptance Inspection Equipment for Non Electronic Components, Design, Testing and Approval, of"

PAGE 2

2.1.1: Specifications and Standards:

Under Product and Packaging drawing 9352634 delete title and substitute "Projectile, 155MM, M825, Metal Parts Assembly".

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2.1.1: Under Inspection Equipment drawings make the following changes:

"T9280354 - Eddy Current Standard, XM825 Base" to:

"9280354 - Eddy Current Standard, M825 Base" and

"T9280355 - Envelope drawing Eddy Current Equipment, XM825 Base" to:

"9280355 - Envelope drawing Eddy Current Equipment, M825 Base"

Change: "9280427 - Envelope Drawing - M825 Base

9280428 - Ultrasonic Standard Drawing, M825 Base"

To: "9280427 - Envelope Drawing - M825A1 Base

9280428 - Ultrasonic Standard Drawing, M825A1 Base".

2.2: Under "American Society for Testing and Material (ASTM)" Add the following:

"ASTM Method E-18-Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials"

PAGE 3

3.2.3: Delete in its entirety and replace with the following:

"Each body (after application of the welded overlay band) shall be austenitized, oil quenched and tempered uniformly to meet the mechanical properties specified on the applicable drawing. No work involving plastic flow is permitted after tempering unless followed by a stress relief. (see 6.12)."

Add the following paragraphs:

"3.3.1.1 Minimum melting of steel. Upon removal of the rotating band material, the steel band seat shall demonstrate visual evidence of minimum melting due to gilding metal adhesion during application of the rotating band. Visual comparison standards will be provided by the Design Agency.

3.3.1.2 Subsurface defects. There shall be no evidence of either intergranular penetration of gilding metal or weld start-stop cratering when the body band seat is turned down to a 5.875 plus or minus 0.010 inch diameter."

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3.3.2: Delete and add the following:

"3.3.2 Rotating band seat. Body material (steel) under the rotating band shall not be visible in the cannelure . diameter.

3.3.2.1 Process control. Body material (steel) under the rotating band shall not extend above the finished bourrelet surface. A process requirement that will control this feature is depicted on the Body drawing (see Dwg. 9352632). The contractor may, at his option, use a control other than the one defined on the drawing. The manufacturing procedure and the inspection plan for controlling the process shall be forwarded to the responsible technical agency for approval prior to its implementation into production."

PAGE 4

3.4: After base add "(9352629)".

3.4.1: Delete and substitute the following:

"3.4.1 Microstructure. The microstructure of the ogive and base (9352629) shall comply with the microstructure depicted in Figures 1A and 1B or 2. Two examples of acceptable microstructure are shown in Figure 1A and 1B. In Figure 1A, the insoluble phase is well broken up, randomly distributed within a matrix which is essentially unrecrystallized. In Figure 1B the insoluble phase is well broken up and randomly distributed within a matrix which is partially recrystallized. An example of a marginally acceptable microstructure is shown in Figure 2. In this figure the insoluble second phase exhibits some discontinuity and appears to be elongated. An example of an unacceptable microstructure is shown in Figure 3. The insoluble phase is distributed in a more continuous form (Network) within a matrix which shows a large degree of microsegration (or coring). The microstructure in Figure 3 must be avoided because it is detrimental to the integrity of the material."

3.4.2: First line, after base add "(9352629)".

3.4.2.1: Delete and substitute the following:

"3.4.2.1 Base (9352629) heat treatment verification. Each base (9352629) shall be Rockwell B85 minimum."

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Add paragraph 3.4.2.2:

"3.4.2.2 Base (9378453) heat treatment verification. Each base (9378453) shall be Rockwell C32 minimum."

3.4.3: Second line, after base, add "(9352629)".

3.6: Delete in its entirety and replace with the following:

"Each base and base adapter shall be austenitized, oil quenched and tempered uniformily to meet the mechanical properties specified on the applicable drawing (see 6.12). No work involving plastic flow is permitted after tempering unless followed by a stress relief."

Add paragraph 3.6.1:

"3.6.1 Verification of base (9378453) heat treatment (special defect). After final machining and prior to phosphate coating, each base (9378453) shall be tested for heat treatment. Bases failing to comply will be rejected. All bases passing acceptance will be stamped in accordance with the applicable drawing. Testing shall be in accordance with 4.5.1.3."

3.7: Delete "XM825" and substitute "M825".

3.8.1: Second line, insert "microstructure," between "process" and "and".

Add paragraph 3.12 as follows:

"3.12 Protective finish. Details of the proposed procedure for zinc phosphate coating, anodic coating to be used by the contractor shall be submitted in writing to the cognizant engineering agency and written approval received prior to initiation of production. The proposed procedure shall include materials, process and equipment to be used by the contractor, detailed method of control including limits for time, temperature, concentration and all other information pertinent to the processes. Approval of process, materials, or equipment implies no guarantee of acceptance of the results obtained in use."

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PAGE 6

4.3.1: Add the following:

"Base	9378453	26
Base (prior to prime and paint)	9378453	5"

PAGE 7

Table I. First article inspection. Sheet 1 of 3, delete in its entirety and substitute the following:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)
AMENDMENT 5

PARAGRAPH	TITLE: Metal Parts Assembly		SHEET 1 OF 3		DRAWING NUMBER: See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ogive				
	(Dwg. 10542916)				
	Examination for defects	5	-	3.1	4.4.2.1
	Protective finish test	1	-	3.1	4.5.10
	Yield strength & percent elongation	2 Specimens	-	3.4.2	4.5.1.5
	Microstructure	1	-	3.4.1	4.5.7
	Forging	-	-	3.8.1	-
	Obturator				
	(Dwg. 10542907)				
	Examination for defects	5	-	3.1	4.4.2.2
	Base (prior to prime & paint)				
	(Dwg. 9378453)				
	Examination for defects	5		3.1	4.4.2.11
	Metal defective	5		3.5.1	4.4.2.11
	Yield strength & percent elongation	5		3.6	4.4.2.11
	Heat treatment verification	5		3.6.1	4.4.2.11, 4.5.1.3
	Base				
	(Dwg. 9378453)				
	Examination for defects	5		3.1	4.4.2.12
	Salt spray	1		3.1	4.5.10

NOTES:

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PAGE 8

Table I. First article inspection. Sheet 2 of 3, delete in its entirety and substitute the following:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER See below
1	Metal Parts Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Adapter, Base (Dwg. 9352630)				
	Examination for defects	5	-	3.1	4.4.2.7
	Yield strength & percent elongation	2 Specimens	-	3.1	4.5.1.6
	Base (Dwg. 9352629)				
	Examination for defects	5		3.1	4.4.2.6
	Yield strength & percent elongation	2		3.4.2	4.5.1.5
	Protective finish test	1		3.1	4.5.10
	Microstructure	1		3.4.1	4.5.7
	Forging	-		3.8.1	-
	Metal defective (prior to drilling wrench slots)	5		3.5.1	4.5.4
	Verification of heat treat (prior to protective finish)	5		3.1	4.5.13
	Base and Adapter Assembly (Dwg. 9352631)				
	Examination for defects	5		3.1	4.4.2.8
	Air leak test	5		3.1	4.5.9
	Protective finish test	1		3.1	4.5.10
	Body (Dwg. 9352532)				
	Examination for defects	5	-	3.1	4.4.2.4
	Yield strength & percent elongation	2	-	3.2.3	4.5.1.4

NOTES:

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Delete paragraph 4.4.1, 4.4.1.1 and 4.4.1.2 in their entirety and substitute the following:

"4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of projectiles shall contain:

a. Bodies from not more than one forging manufacturer.

b. Ogives from not more than one forging manufacturer.

c. Bases (9378453) from not more than one forging manufacturer.

d. Base and adapter assemblies from not more than one forging manufacturer of each component.

4.4.1.1 Heat treatment lot. A heat treatment lot of bodies, bases (9378453), or base adapters shall consist of bodies, bases or base adapters from one mill heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature and for the same time cycles in one unchanged process. The maximum lot size shall be 2000. If a mill heat exceeds 2000, the mill heat shall be divided into any convenient number of heat treatment lots that would preclude the need for forming miscellaneous heat treatment lots. The heat treatment lot thus constituted shall be the basis for selecting samples for mechanical properties tests. Bodies or bases in each heat treatment lot shall be identified to that particular lot until satisfactory completion of mechanical properties tests.

4.4.1.2 Miscellaneous heat treatment lots. Bodies, bases or base adapters manufactured from small heats of steel, or that have become separated from previously delivered lots may be grouped into a miscellaneous heat treatment lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.1. The maximum lot size shall be 2000."

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4.4.1.3: Delete and substitute the following:

"4.4.1.3 Heat treatment lot (base, 9352629) and ogive. A heat treatment lot of ogives and bases shall consist of components manufactured from material of the same alloy and temper from one manufacturer. At no time will the lot quantity exceed 4800. A heat treatment lot consisting of a single component configuration shall be heat treated by the same continuous process, in the same equipment, and using the same temperature and time cycles. A heat treatment lot shall be considered to have been processed continuously if the parts are solutionized and quenched in an uninterrupted manner with no more than a ten (10) hour delay for non-equipment related shutdown(s) provided normal furnace operation temperatures are maintained. If the heat treat process is interrupted to process different components, different lot numbers must be assigned to the parts produced before and after the interruption. Each heat treat lot shall be aged in the same furnace at the same time."

PAGE 11

4.4.1.4: In the title after base add "(9352629)".

4.4.2.a: Add the following to the end of the last sentence:

"or MIL-STD-1235, Inspection Level II where applicable."

PAGE 13

4.4.2.1:

Add "(c)" after defect category 118 and 119.

PAGE 14

4.4.2.1:

Add the following note to the notes block:

"(c) These inspections are not required when the body and ogive are supplied as an assembled unit."

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PAGE 16

Add the following major defect:

"103	Runout of band seat to bourrelets		4.4.2.d	Gage"

PAGE 17

4.4.2.4:

Major 101, Band seat metal defective - change "3.5.2" to "3.3.1, 3.5.2".

PAGE 19

4.4.2.4:

Major 131: Delete in its entirety.

Major 135: Delete "Gage" and substitute "Visual".

PAGE 20

4.4.2.4:

Delete major defect 141.

4.4.2.4: Major 147: Under Examination or Test, delete "&".

Add the following major defect:

"148	Functional runout of body cavity	0.40%	3.2	Gage"

4.4.2.4:

Add "(a)" after defect category 147.

Add the following major defect:

"149	(b) OD of bourrelets, max.	100%	3.1	Gage"

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PAGE 21

Add the following Major defect:

"150	Band seat visible in rotating band cannelures	100%	3.3.2	Visual".

4.4.2.4: Add the following notes to the notes block:

"(a) This inspection is not required when the body and ogive are supplied as an assembled unit.

(b) This inspection is not required when the bourrelet diameter on the Projectile Metal Parts (after paint) is inspected 100%."

PAGE 23

Delete NOTE B and substitute the following:

"(b) Sample size shall be as specified in 4.4.3.3.1 herein."

4.4.2.6:

Major 119: Delete in its entirety.

Major 121: Delete "Gage" and substitute "Visual".

PAGE 25

4.4.2.6: Add the following major examination:

"127	Taper incorrect	0.40%	3.1	Gage"

PAGE 26

4.4.2.7: Delete major category 103 in its entirety.

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PAGE 28

4.4.2.7:

Minor 201, under Examination or Test, add "diameter" after "basic".

Minor 203, under Examination or Test, delete "Diameter on taper" and substitute "Taper incorrect".

Minors 204 and 205, delete "Gage" and substitute "Visual".

PAGE 29

4.4.2.8: Base and adapter assembly.

Critical 1: Add "After assembly and prior to painting."

Major 109: Add "After assembly and prior to painting."

4.4.2.8: Major 104: Delete "Gage" and substitute "Visual".

4.4.2.8: Delete Major 104, 106 and 107 in their entirety.

PAGE 31

4.4.2.9: Major category

102: Add "(c)" after 100% in AQL column.

Add the following note to the notes block:

"(c) An option AQL of 0.40% applies when this defect is inspected 100% on the body before painting (see para 4.4.2.4, major defect 149)."

4.4.2.9: Add the following major categories:

"107	Lifting plug or gasket missing	-	100%	3.1	Visual
108	Lift plug detorque	-	0.40%	3.1	Gage"

4.4.2.9: Minor category

208: Delete "& fuze".

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PAGE 33

4.4.2.10: Major category Delete "None defined" and add the following:

"101	Support cover missing (8 places) when applicable	100%	3.1	Visual".

PAGES 33a and 33b

4.4.2.11: Add this paragraph (2 sheets attached).

PAGE 33c

4.4.2.12: Add this paragraph (1 sheet attached).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)
AMENDMENT 5

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9378453
4.4.2.11	Base (prior to prime and paint)				NEXT HIGHER ASSEMBLY 9352634
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special 1	Heat treatment verification		100%	3.6.1	4.5.1.3
Major 101	Metal defective		100%	3.5.1	4.5.4.2 (UT)
102	Yield strength and percent elongation		(a)	3.6	4.5.1.7
103	Length from shoulder to undercut at aft end of thread		0.40%	3.1	Gage
104	Depth to basic diameter in cavity radius		0.40%	3.1	Gage
105	Length from shoulder to forward end		0.40%	3.1	Gage
106	Pitch diameter of thread, min.		0.40%	3.1	Gage
107	Major diameter of thread, min.		0.40%	3.1	Gage
108	Diameter of O-ring groove		0.40%	3.1	Gage
109	True position of O-ring groove		0.40%	3.1	Gage
110	Width of O-ring groove		0.40%	3.1	Gage
111	Diameter at forward edge of thread		0.40%	3.1	Gage
112	Angle of thread lead and follow chamfer, max.		0.40%	3.1	Gage
113	Surface finish of O-ring groove and shoulder		0.40%	3.1	Visual/Comparison Blocks
113	Straightness of shoulder		0.40%	3.1	Gage

NOTES: (a) 4.5.1.6

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9378453
4.4.2.11	Base (prior to prime and paint)				NEXT HIGHER ASSEMBLY 9352634
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
115	Length from shoulder to basic boattail diameters (2 places)		0.40%	3.1	Gage
116	Runout of boattail (2 places)		0.40%	3.1	Gage
117	Thickness of radius webb		0.40%	3.1	Gage
118	Length of boattail		0.40%	3.1	Gage
Minor					
201	Inside diameter of boattail		0.65%	3.1	Gage
202	Runout of boattail inside diameter		0.65%	3.1	Gage
203	Boattail-webb radius		0.65%	3.1	Gage
204	Width of wrench slots, max.		0.65%	3.1	Gage
205	Depth of wrench slots, max.		0.65%	3.1	Gage
206	True position of wrench slots		0.65%	3.1	Gage
207	Any feature or operation missing		0.65%	3.1	Gage
208	Surface finish incorrect, except O-ring groove and shoulder		0.65%	3.1	Visual/Comparison Blocks
209	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64008 (AR)
AMENDMENT 5

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9378453
4.4.2.12	Base				NEXT HIGHER ASSEMBLY 9352634
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major					
101	Threads damaged		100%	3.1	Visual
102	Salt spray		(a)	3.1	4.5.10
Minor					
201	Protective coating missing, incomplete or damaged		0.65%	3.11	Visual
202	Paint on threads, O-ring groove or shoulder		0.65%	3.1	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: (a) Sample size shall be specified in 4.4.3.3.1.

AMSMC Form 1570, 1 Feb 85 — Replaces DRSTR-QA (D) Form 160, 1 Aug 83, which may not be used.

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PAGE 34

4.4.3.3.1: Delete in its entirety and substitute the following:

"4.4.3.3.1 Base (9352629) and ogive. First article bases and ogives shall withstand the salt spray test as prescribed in paragraph 3.10.1.2 of MIL-A-8625. Production lot bases and ogives shall comply with paragraph 3.10.1.2 of MIL-A-8625 except exposure shall be for 72 hours and show no more than 3 corrosion spots and/or pits on the significant surface. The significant surface on the ogive as prescribed in MIL-A-8625 may be marked by tracing the outline of a 5 inch by 6 inch sheet of paper onto the ogive."

4.4.3.3.3: Delete "XM825" and substitute "M825".

Add paragraph 4.4.3.3.4 as follows:

4.4.3.3.4 Ogive. First article ogives shall withstand the salt spray test as prescribed in para 3.10.1.2 of MIL-A-8625. Production lot ogives shall comply with para 3.10.1.2 of MIL-A-8625 except exposure shall be for 72 hours and show no more than 3 corrosion spots and/or pits on the significant surface. the significant surface as prescribed in MIL-A-8625 may be marked by tracing the outline of a 5 inch by 6 inch sheet of paper onto the ogive."

PAGE 35

* 4.5.1.2a: Delete the first sentence and substitute the following:

"The surface of the body in the area to be hardness tested shall be prepared by machining or grinding a flat, or machining to a controlled diameter of sufficient depth, 0.015 inches minimum to remove scale and get below the decarburization zone."

4.5.1.2: Add the following to the end of paragraph a:

"Manual grinding is not permitted."

PAGE 36

4.5.1.3: Delete and substitute the following:

"4.5.1.3 Verification of heat treat of base. The test shall be made using procedures as prescribed in ASTM E-18. The readings shall be taken on the surface indicated on the applicable drawing."

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Delete paragraph 4.5.1.5 in its entirety and substitute:

"4.5.1.5 Yield strength and percent elongation of ogive. The sample size for the tension test shall be one component from each 1200 parts or portion thereof from each heat treatment lot. Two longitudinal test specimens conforming to the largest obtainable round specimen as prescribed by ASTM Method E-8 shall be taken from the sidewall of each sample. The test specimens shall be taken from the midwall at a location where the gage length center of the specimen is approximately 1 inch from the finished rear face of the body thread end of the ogive, 180 apart. The criteria for performing the tests shall be as prescribed by ASTM Method E-8. Percent elongation shall be recorded to the nearest tenth percent. Failure of any specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the quantity represented by the sample."

Add 4.5.1.7 as follows:

"4.5.1.7 Hardness (base, 9378453). In order to determine uniformity of heat treatment, each base after heat treatment and before final machining shall be subjected to a hardness test with a Brinell Hardness Tester or any other method approved by the procuring activity. The test shall be performed in accordance with ASTM Method E-103 and the following:

a. The surface of the base in the area to be hardness tested shall be prepared by machining or grinding a flat of sufficient depth, .015 inches minimum, to remove scale and get below the decarburization zone. Flats and impressions shall be completely removed by final machining. manual grinding is not permitted.

b. Hardness tester periodic verification shall be performed using Brinell test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically recorded.

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d. The hardness test shall use a standard 10MM Brinell ball. The hardness tester shall determine hardness in accordance with the following test sequence. First, the tester shall clamp the base to be tested with a 10,000 -500 lb. clamping load. Second, apply a minor load of approximately 10% of the 3000 kg of total test load. Third, apply the total test load. The clamping load will not be exceeded when the test load is applied. Fourth, relax the total test load back to the minor load. Fifth, automatically measure this penetrator travel from initial minor load position to new minor load position following application of the total test load. Sixth, display hardness and release the base."

Add paragraph 4.5.1.8 as follows:

"4.5.1.8 Yield strength and percent elongation of base. One of the hardest and one of the softest bases from each standard heat treatment lot or two of the hardest and two of the softest bases from each miscellaneous heat treatment lot as determined by the hardness test in 4.5.1.7 shall be subjected to this test. Samples for mechanical property testing shall be selected after heat treatment of the entire heat treatment lot. From each base selected, one test specimen shall be taken transversely from near the center of the base webb, and two specimens shall be taken longitudinally 180 apart from the boattail forward corner area. The specimens shall conform to the largest obtainable round type diameter specimen as prescribed by ASTM Method E-8. If a .113 inch diameter cannot be machined from the longitudinal area of the base heat treat configuration, one rough forged base of the same mill heat of steel shall be included in each heat treatment lot and two longitudinal specimens shall therefrom be machined. The criteria for performing the tests shall be as prescribed by ASTM Method E-8. Failure of any longitudinal specimen to comply with the yield strength and elongation requirements, or failure of any transverse specimen to comply with the yield strength requirements of the applicable drawing shall be cause for rejection of the heat treatment lot. At the option of the contractor, the entire lot may be selected and tested from a segment of the heat treat lot having a tighter hardness range than the heat treatment lot as originally sampled. If an acceptable hardness range can be established for this heat treatment lot, bases falling within this range will be accepted. All other bases will be rejected. Rejected bases may be reheat treated and retested."

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4.5.3.1: Delete paragraph and substitute the following:

"4.5.3.1: Inspection of band seat.

4.5.3.1.1 Welder qualification sampling. Initially the first five banded bodies after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to the tests outlined below. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been found acceptable. These bodies shall have the band removed by machining and chemically etching without damaging the band seat metal (steel) (see 6.7). These bodies shall be subjected to the following tests:

a. Visual examination for compliance with 3.3.1.1 minimum melt requirement.

b. Visual examination for compliance with 3.5.2 metal defects, using the technique described in 4.5.3.1.3.

c. Turn down band seat to the diameter specified in 3.3.1.2 and again visually inspect in accordance with 4.5.3.1.3. Any defect indication per 3.3.1.2 at this depth is a reject.

d. Should any sample for any of the above tests fail to comply with the minimum acceptance requirements, the quantity of bodies represented by that sample shall be rejected.

4.5.3.1.2 Post qualification sampling.

4.5.3.1.2.1 Etched band seat examination. Reduced sampling of one in each 8 hours production or fraction thereof per welding machine in operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been found acceptable (see 6.9). One body per welder will be randomly selected and inspected for the following:

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a. Visual examination for compliance with 3.3.1.1, minimum melting. The body selected for this test need not be the test sample having the lower rotating band iron content when tested in accordance with 4.5.5.

b. Visual inspection for compliance with 3.5.2, metal defects, using the technique described in 4.5.3.1.3. If the sample fails to comply with 3.5.2 then the band seat will be turned down to the diameter specified in 3.3.1.2 and again visually inspected in accordance with 4.5.3.1.3.

c. Should any sample fail to comply with the minimum acceptance requirements, the quantity of bodies represented by that sample shall be rejected.

d. Two consecutive failures when tested for a, or two consecutive failures when tested for b, for any welding machine will mandate the return to the original welder qualification sampling of paragraph 4.5.3.1.1. However, reduced sampling of one in each 8 hours production or fraction thereof may be implemented after the first five banded bodies have been found acceptable.

4.5.3.1.2.2 Sub-surface defect examination. The body having the highest acceptable iron content rotating band when tested per paragraph 4.5.5 during this 8 hour post qualification sampling shall be tested for compliance with 3.3.1.2. This body will have its band seat turned down to the diameter specified in paragraph 3.3.1.2 and visually inspected per paragraph 4.5.3.1.3. Failure of the body to comply with the specified requirements is cause for rejection of all bodies having bands welded by the same welder during the same sampling period. Failure of any body shall also require that the body exhibiting the highest iron content from each welder be similarly tested. Failure of any body is cause for rejection of the quantity from each discrete welder represented by the sample. Two consecutive failures on any welding machine will mandate the return to the original welder qualification sampling of paragraph 4.4.3.1.1. However, reduced sampling of one in each 8 hours production or fraction thereof may be implemented after the first five banded bodies have been found acceptable.

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4.5.3.1.3 Magnetic particle inspection. Magnetic particle inspection shall be as specified in MIL-I-6868 for wet fluorescent method with indirect magnetization by the continuous method, initially after inducing a circular magnetic field and again after inducing a longitudinal magnetic field.

4.5.3.1.4 Retest. If the sample selected in 4.5.3.1.2.2 fails to comply with the requirements, then a minimum of 10% of the shell from the failed welder may be randomly selected for retest. These parts will be tested in accordance with 4.5.3.1.2.2. Only one retest per welder is allowed. Failure of any part will mandate the rejection of the product represented by the sample.

4.5.3.2: Add the following after the first sentence:

"At the option of the contractor, this test may be performed prior to machining of the rear counterbore, keyway/threads."

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Delete paragraph 4.5.4 in its entirety and substitute the following:

"4.5.4 Metal defective test.

4.5.4.1 Base (9352629). The base shall be inspected for metal defects using the Eddy Current method. Bases failing to comply with the requirements of 3.5 shall be classed defective and removed from the lot. All bases passing this test shall be stamped in accordance with their applicable drawing. The standards and test method shall be in accordance with drawings 9280354 and 9280355.

4.5.4.2 Base (9378453). Every base shall be ultrasonically inspected for metal defects. Standards and test methods shall be in accordance with drawings 9280427 and 9280428 and with MIL-A-70625. Stamp letter "U" after acceptance as indicated on the base drawing.

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4.5.4.2.1 Calibration of ultrasonic equipment. As a minimum the adequacy of the inspection process shall be checked by using the standards at the start and finish of each production shift or change in operator, as well as four (4) intervals during production. If the inspection process is determined to be inadequate, the condition shall be corrected and all items inspected since the last acceptable inspection check shall be reinspected."

4.5.5: Delete the next to last sentence.

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Delete paragraph 4.5.7 in its entirety and substitute the following:

"4.5.7 Microstructure examination (ogive) and base, (9352629). One ogive and one base forging from each heat treatment lot shall be selected for microstructure examination. The metallographic specimen shall be etched with Keller's reagent and all areas examined under a microscope/metallograph at a minimum magnification of 250 diameters. Any evidence of the insoluble phase generally occurring in a semi-continuous network form shall be cause for rejection of the lot represented by the sample. Microstructural preparation and examination shall be performed by a metallographer or metallurgical engineer familiar with aluminum alloys."

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Delete title, "Figure 2 Unacceptable Microstructures (250X)"

Delete "a" and substitute "Figure 2. Marginally Acceptable Microstructure (250X)"

Delete "b" and substitute "Figure 3. Unacceptable Microstructure (250X)"

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5.1.1, 5.1.2, and 5.1.3: Delete in their entirety.

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5.2: Delete and substitute the following:

"5.2 Packaging, packing and marking."

5.2.2: Delete and substitute the following:

"5.2.2 Packaging, Packing and Marking of Bases shall comply with the following:"

Add the following paragraphs after paragraph 5.2.2:

5.2.2.1 Overseas shipments. Bases shall be semi-bulk or single item packaged, moisture proof packaged (barrier bag only), packed with cavity facing downward, marked, and palletized in accordance with MIL-STD-1169, Level A. Bases shall be separated or partitioned (if bulk) in the unit package to protect the paint surfaces and shall include marking to assure that the base is shipped or stored with the cavity facing downward.

5.2.2.2 Domestic shipments. Bases shall be semi-bulk or single item packaged, packed with the cavity facing downward, and marked in accordance with MIL-STD-1169, Level B. Bases shall be separated or partitioned (if bulk) in the unit package to protect the paint surfaces and shall include marking to assure that the base is shipped or stored with the cavity facing downward. Palletization is optional."

6.1: Intended use: Change "XM825" to "M825/M825A1".

6.3 Delete "ARRADCOM, Attn: DRDAR-QAR-I, Dover, NJ 07801" and substitute "AMCCOM, Attn: SMCAR-QAR-I(D), Picatinny Arsenal, NJ 07806-5000."

6.4 Delete "ARRADCOM, Attn: DRDAR-QAR-Q, Dover, NJ 07801" and substitute "AMCCOM, Attn: SMCAR-QAR-Q(D), Picatinny Arsenal, NJ 07806-5000."

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6.5 Delete the last two lines, and substitute:

"AMCCOM, Attn: AMSMC-QAR-Q(D), Picatinny Arsenal, NJ 07806-5000 and PMCAWS, Attn: SMCAR-CCH-P, Picatinny Arsenal, NJ 07806-5000."

6.6 Delete "XM825" and substitute "M825".

6.9 Delete "ARRADCOM, Attn: DRDAR-LCU-M, Dover, NJ 07801" and substitute "ARDEC, Attn: SMCAR-CCH-P, Picatinny Arsenal, NJ 07806-5000."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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